

1. Administrative & Study Identification

Full Study Title: Study of Acalabrutinib in Chinese Adult Subjects With Relapsed or Refractory Mantle Cell Lymphoma, Chronic Lymphocytic Leukemia or Other B-cell Malignancies **Short Title/Acronym:** Acalabrutinib in Chinese R/R MCL and CLL **Protocol Number:** D8220C00007 **NCT Number:** NCT03932331 **Sponsor Name:** AstraZeneca **Investigational Product:** Acalabrutinib (Calquence) **Phase of Development:** Phase I/II **Trial Status:** Active, not recruiting **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, and pharmacokinetics of acalabrutinib (including Adverse Events, Serious Adverse Events, and Area Under the Plasma Concentration-Time Curve [AUCinf and AUC0-12]) in Chinese adult patients with relapsed or refractory (R/R) Mantle Cell Lymphoma (MCL), Chronic Lymphocytic Leukemia (CLL), or other B-cell malignancies. **Secondary Objectives:** To evaluate clinical efficacy, specifically Best Overall Response (BOR) and Overall Response Rate (ORR) assessed by the investigator, as well as Duration of Response (DoR), Progression-Free Survival (PFS), Overall Survival (OS), and Time to Response (TTR).

2.2 Background & Rationale To determine whether the safety, tolerability, pharmacokinetics, and efficacy of acalabrutinib in a Chinese patient population are comparable to previously reported global phase 3 trial results that included predominantly Caucasian patients with R/R CLL and MCL. This is particularly important to address the notable differences in the frequency of common genetic mutations and chromosomal abnormalities between Asian and European patients with CLL/MCL.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional, open-label, two-part study (Phase 1 and Phase 2) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **NS:** 105 participants **Number of Sites:** 20 clinical sites in China **Estimated Study Start:** 2019 **Estimated Primary Completion:** December 2023 (Data cutoff for primary analysis)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Capable of giving signed informed consent which includes compliance with the requirements and restrictions listed in the informed consent form (ICF) and in this protocol. 2. Chinese adults ≥ 18 years of age at the time of study entry. 3. Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 2 . 4. Adequate hematological and organ function. 5. Presence of radiographically measurable lymphadenopathy or extranodal lymphoid malignancy. 6. Pathologically confirmed MCL, with documentation of chromosome translocation t(11;14) (q13;q32) and/or overexpression of cyclin D1 in association with other relevant markers (e.g., CD5, CD19, CD20, PAX5). Disease had relapsed after or been refractory to previous treatment. 7. Diagnosis of CLL that meets published diagnostic criteria. Must have received ≥ 1 prior systemic therapies for CLL. 8. Active disease per iwCLL 2018 criteria that requires treatment (CLL only). 9. Other relapsed/refractory B-cell malignancies without standard of care (Phase 1 only).

4.2 Exclusion Criteria 1. Prior malignancy, except for adequately treated basal cell or squamous cell skin cancer, in situ cervical cancer, or other cancer from which the subject had been disease free for ≥ 2 years or which would not have limited survival to < 2 years. 2. Significant cardiovascular disease. 3. Known central nervous system involvement of lymphoma/leukemia or leptomeningeal disease. 4. Known history of HIV, serologic status reflecting active hepatitis B or C infection. 5. Major surgery within 4 weeks before the first dose of study drugs. 6. Uncontrolled autoimmune hemolytic anemia or idiopathic thrombocytopenia purpura. 7. Required or received anticoagulation with warfarin or equivalent vitamin K antagonist (e.g., phenprocoumon). 8. Prior exposure to a BCR or BCL-2 inhibitor. 9. Use of a strong inhibitor or inducer of CYP3A. 10. Breastfeeding or pregnant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of acalabrutinib 100 mg capsule orally on Day 1 Cycle 0, followed by a 2-day washout period. Subsequently, participants received acalabrutinib 100 mg capsule orally twice daily in 28-day cycles. **Route of Administration:** Oral **Duration of Treatment:** Continuous treatment in 28-day cycles until disease progression (PD) or until specific treatment discontinuation criteria/unacceptable toxicities are met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for oncology patients. **Prohibited:** Anticoagulation with warfarin or equivalent vitamin K antagonists (e.g., phenprocoumon); use of a strong inhibitor or inducer of CYP3A.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent,
Medical History, Baseline Tumour Assessment | | **Baseline** | Day 0
| Cycle 0 single dose, 2-day washout, start of 100 mg twice daily
| | **Interim** | End of Cycles 2, 4, 6 | Safety Labs, Efficacy
Assessment (Response evaluations) | | **Long-term Follow-up** | Every
12 / 24 Weeks | Response evaluations every 12 weeks through Cycle
25; every 24 weeks thereafter | | **End of Study** | At progression |
Final Vital Signs, Exit Interview, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Phase I:** Number of Participants With Adverse Events (AEs) and Serious Adverse Events (SAEs). 2. **Phase I:** Area Under the Plasma Concentration-Time Curve From Zero to Infinity (AUC_{inf}) of Acalabrutinib and ACP-5862 Post Single Dose of Acalabrutinib (determined using non-compartmental method). 3. **Phase I:** Area Under the Plasma Concentration-Time Curve From Zero to 12 Hours (AUC₀₋₁₂) of Acalabrutinib and ACP-5862 Post Single Dose of Acalabrutinib (determined using non-compartmental method).

7.2 Secondary & Exploratory Endpoints 1. **Phase I:** Best Overall Response (BOR) Assessed by Investigator in Participants With R/R CLL. 2. **Phase I:** Best Overall Response (BOR) Assessed by Investigator in Participants With R/R MCL. 3. **Phase II:** Overall Response Rate (ORR) Assessed by Investigator. 4. **Other secondary endpoints:** Duration of Response (DoR), Progression-Free Survival (PFS), and Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard reporting of Treatment-Emergent Adverse Events (TEAEs) graded by severity. An AE leading to discontinuation is action taken as "Drug Permanently Discontinued." **Serious Adverse Events (SAEs):** Reported per institutional guidelines for events that are fatal, life-threatening, require hospitalization, result in persistent/significant disability, congenital anomaly, or represent an important medical event. **Data Monitoring Committee (DMC):** Blinded Independent Central Review (BICR) for efficacy endpoints.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy-evaluable population (All enrolled patients receiving at least one dose of study drug). **Sample Size**

Justification: Powered to evaluate ORR; ORR and the corresponding 95% 2-sided CI were presented based on the Clopper-Pearson exact method. **Handling of Missing Data:** The Duration of Response (DoR), PFS, and OS were calculated using the Kaplan-Meier technique. Pharmacokinetics (AUCinf, AUC0-12) assessed via non-compartmental method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring per oncology study guidelines. **Audit Trail:** Robust centralized review of radiological and clinical responses via BICR to ensure unbiased evaluation.

11. Study Identifiers & Governance

Primary ID: D8220C00007 **Registry Number:** NCT03932331 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Study of Acalabrutinib in Chinese Adult Subjects With Relapsed or Refractory Mantle Cell Lymphoma, Chronic Lymphocytic Leukemia or Other B-cell Malignancies **Official Regulatory Title:** A Phase 1/2 Open Label, Multi-center Study to Assess the Safety, Tolerability, Pharmacokinetics and Clinical Efficacy of Acalabrutinib in Chinese Adult Subjects With Relapsed or Refractory Mantle Cell Lymphoma, Chronic Lymphocytic Leukemia or Other B-cell Malignancies

12. Clinical Framework (Oncology Specific)

Primary Condition: Phase I: Relapsed or Refractory B-cell Malignancies, Phase II Cohort A: Relapsed or Refractory Mantle Cell Lymphoma, Phase II Cohort B: Relapsed or Refractory Chronic Lymphocytic Leukemia **Diagnosis Threshold:** Histologically confirmed MCL or CLL with ≥ 1 prior line of systemic therapy **Medical Methodology:** Targeted therapy (BTK inhibitor) **Optimization Target:** Safety assessment, Pharmacokinetics, Disease Control, and Overall Response Rate (ORR) maximization

13. Study Procedures & Methodology

Management Pathway: Continuous 28-day treatment cycles until disease progression **Education Protocol (HCP):** Standard oncology protocol training & AE management **Education Protocol (Patient):** Oral medication adherence and AE self-reporting **Quality Audit**

Mechanism: Blinded Independent Central Review (BICR) of imaging/scans

14. Observation & Follow-Up Schedule

Total Duration: Median treatment duration of ~14 to 19.4 months (until progression) **Onsite Visit Cadence:** Cycles 2, 4, 6, then every 12 weeks up to Cycle 25, then every 24 weeks **Remote Monitoring Frequency:** Continuous AE reporting **Checklist Review Interval:** Every 28-day cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					